ONLINE QUIZ SYSTEM

Software Requirements Specification (SRS)

Course: Software Engineering
Class: 5th Semester – Section H
Team: 13
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Document Control

Version	Date	Description
1.0	02-11-2026	Initial consolidated Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Online Quiz System. This document describes the system's features, users, constraints, interfaces, security requirements and expected behaviour.

It serves as a reference for the development, testing and validation of the Online Quiz System.

1.2 Problem Statement

Traditional quiz and examination processes may involve manual question distribution, answer evaluation and result preparation. These activities can consume time and introduce errors.

The Online Quiz System aims to provide a simple web-based platform through which participants can attempt multiple-choice quizzes and receive automatically calculated results. Administrators can securely manage quiz questions through a separate administrative interface.

1.3 Scope

The Online Quiz System is a browser-based application that supports two primary user roles:

Participant

- Access the quiz start page.
- Enter the required participant details.
- View quiz questions and answer options.
- Navigate between questions.
- Submit answers.
- View the calculated result.

Administrator

- Log in using administrator credentials.
- Add new quiz questions.
- View existing questions.
- Edit questions and answer options.
- Delete questions after confirmation.

The first version focuses on multiple-choice questions (MCQs). Advanced features such as online proctoring, payment integration, AI-generated questions and adaptive testing are outside the current project scope.

1.4 Objectives

The main objectives are:

1. Develop a simple and user-friendly online quiz platform.
2. Automate the evaluation of submitted answers.
3. Reduce the time required for manual evaluation.
4. Provide administrators with question-management functionality.
5. Maintain the integrity and security of quiz data.
6. Provide participants with immediate access to their results after submission.

1.5 Definitions and Abbreviations

Term	Description
OQS	Online Quiz System
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
MCQ	Multiple-Choice Question
API	Application Programming Interface
DBMS	Database Management System
UI	User Interface
UML	Unified Modeling Language
RTM	Requirement Traceability Matrix

1.6 References

- IEEE-style Software Requirements Specification structure.
- Software Engineering course project guidelines.
- Team 13 Online Quiz System project description.

2. Overall Description

2.1 Product Perspective

The Online Quiz System is a web-based application designed using a three-tier architecture.

The three tiers are:

1. **Presentation Layer:** HTML, CSS and JavaScript provide the user interface.
2. **Application Layer:** Node.js and Express.js process requests, implement business logic and expose APIs.
3. **Database Layer:** MySQL stores administrator details, quizzes, questions, answer options, attempts and results.

The frontend communicates with the backend through HTTP-based APIs. The backend communicates with the MySQL database.

2.2 Product Functions

The system provides the following major functions:

- Participant details entry.
- Quiz question display.
- Answer selection and question navigation.
- Quiz submission.
- Automatic score calculation.
- Result display.
- Administrator authentication.
- Question creation, viewing, editing and deletion.
- Input validation and error handling.

2.3 User Classes

Participant: A user who attempts quizzes. Participants require basic web-browsing knowledge.

Administrator: An authorised user responsible for maintaining the quiz question bank. Administrative access requires authentication.

2.4 Operating Environment

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Node.js
Backend Framework	Express.js
Database	MySQL
Communication	HTTP/HTTPS and JSON
Client	Modern web browser
Development Environment	Windows/Linux/macOS

2.5 Design and Implementation Constraints

1. The frontend must be developed using HTML, CSS and JavaScript.
2. The backend must use Node.js and Express.js.
3. MySQL must be used for persistent data storage.
4. Administrative operations must require authentication.

5. Server-side validation must be performed for user inputs.
6. The system requires network connectivity between the browser, backend and database.

2.6 Assumptions and Dependencies

1. An administrator account will be configured for managing questions.
2. Quiz questions and their correct answers will be available in the database.
3. Participants will have access to a supported web browser.
4. The backend and database services will be available while the system is being used.
5. The team will confirm quiz duration, participant information fields, retake policy and scoring rules before implementation.

3. Functional Requirements

Each functional requirement is assigned a unique identifier. These identifiers must remain unchanged in the Test Plan and Architecture & Design Specification.

ID	Functional Requirement
FR-01	The system shall allow a participant to access the quiz start page and view available quiz information.
FR-02	The system shall allow a participant to enter the required participant details before starting a quiz.
FR-03	The system shall display quiz questions and their available answer options.
FR-04	The system shall allow a participant to navigate between questions without losing previously selected answers.
FR-05	The system shall allow a participant to submit the quiz, with confirmation where applicable.
FR-06	The system shall calculate the participant's score using the configured correct answers and scoring rules.
FR-07	The system shall display the participant's result after successful submission and evaluation.
FR-08	The system shall allow an administrator to authenticate before accessing administrative functions.
FR-09	The system shall allow an authenticated administrator to add quiz questions and answer options.

FR-10	The system shall allow an authenticated administrator to edit existing quiz questions and answer options.
FR-11	The system shall allow an authenticated administrator to delete quiz questions after confirmation.
FR-12	The system shall allow an authenticated administrator to view the available question list.
FR-13	The system shall validate required fields and answer configuration before saving a question.

4. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational expectations of the system.

ID	Category	Requirement
NFR-01	Performance	Under normal test conditions, at least 95% of page/API requests should complete within 3 seconds.
NFR-02	Capacity	The system should support at least 50 concurrent participant sessions in the agreed test environment.
NFR-03	Security	Administrator passwords shall be stored using a secure salted password-hashing algorithm.
NFR-04	Access Control	Administrative APIs shall reject unauthenticated and unauthorised requests.
NFR-05	Reliability	Previously selected answers shall remain available during normal navigation within an active quiz attempt.
NFR-06	Usability	The interface shall provide readable content, clear navigation controls and understandable validation messages.
NFR-07	Compatibility	The application should work on current stable versions of Chrome, Edge and Firefox.
NFR-08	Maintainability	Frontend, backend business logic and database-access modules shall be organised separately.
NFR-09	Data Integrity	The server shall validate data and use suitable database constraints and transactions to maintain consistency.
NFR-10	Error Handling	The system shall handle expected application and database errors without exposing sensitive technical information to users.

Note: The performance and concurrency values are proposed project acceptance targets. The team should confirm that they are achievable with the available testing environment and obtain guide approval.

5. Security Requirements

5.1 Security Objectives

SO-01 – Confidentiality: Protect administrator credentials and prevent unauthorised disclosure of restricted information.

SO-02 – Integrity: Prevent unauthorised or invalid modification of questions, answer keys and stored results.

SO-03 – Access Control: Ensure that only authenticated and authorised administrators can perform question-management operations.

SO-04 – Safe Error Handling: Prevent disclosure of passwords, database information, server stack traces and other sensitive implementation details.

5.2 Security Requirements

ID	Security Requirement
SEC-01	Only authenticated administrators shall be allowed to access question-management operations.
SEC-02	Administrator passwords shall be stored as salted password hashes and shall never be stored or logged in plaintext.
SEC-03	The server shall validate client inputs and use parameterised SQL queries to prevent SQL injection.
SEC-04	Authentication credentials and session identifiers shall be protected in transit and configured securely.
SEC-05	The application shall return generic error messages without exposing stack traces, database details or secrets.
SEC-06	Administrative operations shall be restricted to users with the authorised administrator role.

5.3 Security Considerations

The application shall use HTTPS in its deployed environment. Database credentials shall be stored outside the source code. Administrative APIs shall be protected using authentication and authorisation middleware. Correct answers shall not be exposed to participants before quiz submission.

6. Use Case Model

6.1 Actors

Participant: Starts and attempts a quiz, submits answers and views the result.

Administrator: Logs in and manages quiz questions.

6.2 Use Cases

Use Case ID	Use Case	Actor
UC-01	Start Quiz	Participant
UC-02	Enter Participant Details	Participant
UC-03	Attempt Quiz	Participant
UC-04	Navigate Questions	Participant
UC-05	Submit Quiz	Participant
UC-06	Calculate Score	System
UC-07	View Result	Participant
UC-08	Administrator Login	Administrator
UC-09	Add Question	Administrator
UC-10	View Questions	Administrator
UC-11	Edit Question	Administrator
UC-12	Delete Question	Administrator

6.3 Use Case Descriptions

UC-01: Start Quiz

- Primary actor: Participant.
- Precondition: Quiz is available.
- Main flow:
    1. Participant opens the quiz start page.
    2. System displays quiz information.
    3. Participant enters the required details.
    4. System validates the details.
    5. System starts the quiz.

- Postcondition: A quiz attempt is created.

UC-02: Attempt Quiz

- Primary actor: Participant.
- Precondition: A quiz attempt has started.
- Main flow:
    1. System displays a question and its options.
    2. Participant selects an answer.
    3. Participant navigates to another question.
    4. System retains the selected answers during the attempt.
- Postcondition: The participant's responses are available for submission.

UC-03: Submit Quiz

- Primary actor: Participant.
- Precondition: A quiz attempt is active.
- Main flow:
    1. Participant selects the submit option.
    2. System requests confirmation if configured.
    3. System records the submission.
    4. System calculates the score.
    5. System makes the result available.
- Postcondition: The attempt is marked as submitted and evaluated.

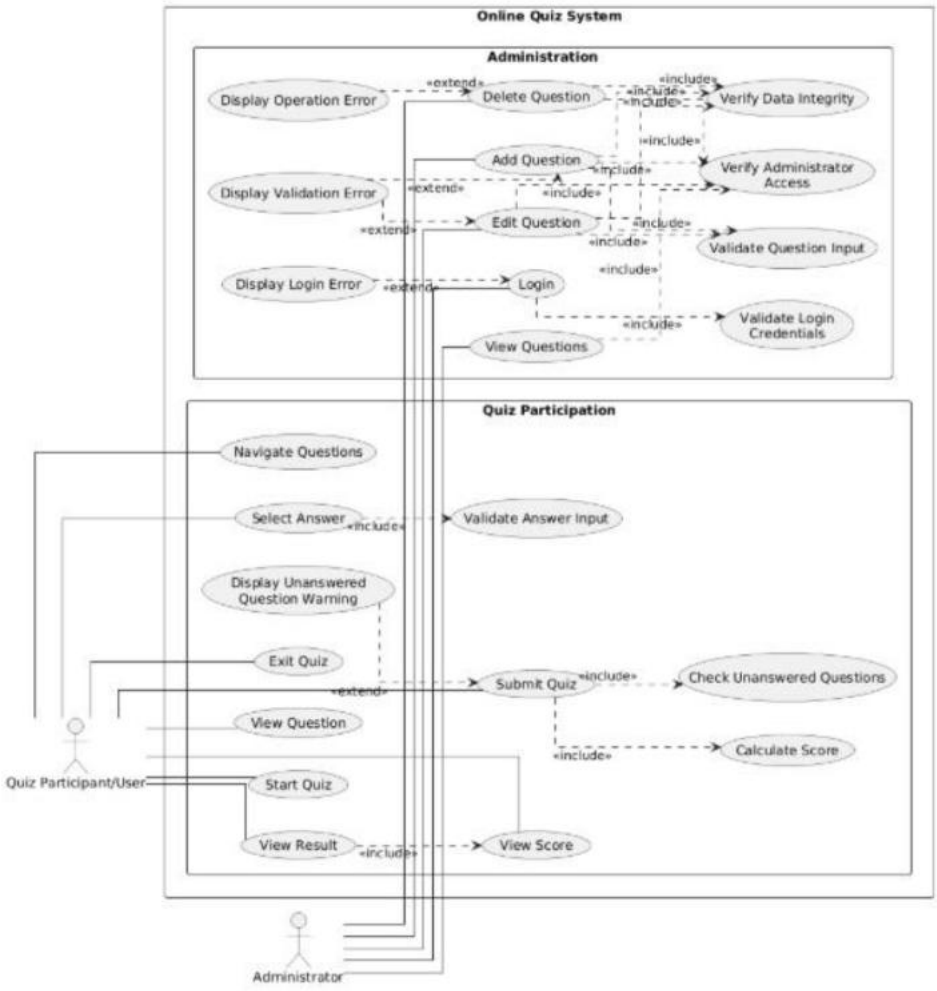
UC-04: Administrator Login

- Primary actor: Administrator.
- Precondition: Administrator account exists.
- Main flow:
    1. Administrator opens the login page.
    2. Administrator enters credentials.
    3. Server verifies the credentials.
    4. System grants access if authentication succeeds.
- Alternative flow: Invalid credentials are rejected with a generic error message.
- Postcondition: An authenticated administrator session is established.

UC-05: Manage Questions

- Primary actor: Administrator.
- Precondition: Administrator is authenticated.
- Main flow:
    1. Administrator opens question management.
    2. Administrator selects add, view, edit or delete.
    3. System validates the request.
    4. System updates the database.
    5. System displays a confirmation.
- Postcondition: The question bank reflects the authorised operation.

6.4 Use Case Diagram



The diagram must show:

- Participant and Administrator actors.

- Participant use cases: Start Quiz, Enter Details, Attempt Quiz, Navigate Questions, Submit Quiz and View Result.
- System use case: Calculate Score.
- Administrator use cases: Login, Add Question, View Questions, Edit Question and Delete Question.
- An include relationship from Submit Quiz to Calculate Score.
- An authentication precondition for all administrator management use cases.

7. External Interface Requirements

7.1 User Interface

The participant interface shall provide quiz information, question text, answer options, navigation controls, submission and result display.

The administrator interface shall provide a login page and question-management screens.

7.2 Software Interface

The frontend shall communicate with the Express.js backend through JSON-based REST APIs. The backend shall access MySQL through a suitable Node.js database driver.

7.3 Communication Interface

The system shall use HTTP during local development and HTTPS in the deployed environment.

8. Data Requirements

The proposed database shall contain the following main entities:

Entity	Important attributes
Administrator	admin_id, username, password_hash, role
Quiz	quiz_id, title, description, status
Question	question_id, quiz_id, question_text, marks
Option	option_id, question_id, option_text, is_correct
Attempt	attempt_id, quiz_id, participant details, started_at, submitted_at, score
Response	response_id, attempt_id, question_id, selected_option_id

Primary keys and foreign keys shall be used to maintain relationships between records.

9. Acceptance Criteria

The system will be considered functionally acceptable when:

1. Participants can start and attempt an available quiz.

2. Selected answers are retained during question navigation.
3. The system submits and evaluates quiz attempts correctly.
4. Participants can view their calculated results.
5. Administrators can authenticate and manage questions.
6. Unauthenticated users cannot perform administrative operations.
7. Invalid input is rejected without corrupting stored data.
8. The agreed non-functional and security requirements have been tested.

10. Requirement Traceability

The functional requirements FR-01 to FR-13, non-functional requirements NFR-01 to NFR-10 and security requirements SEC-01 to SEC-06 shall be referenced consistently in the Test Plan and Architecture & Design Specification.

Changes to requirement IDs or definitions must be reflected in all three documents.